SIMATS ENGINEERING
ASSESSMENT TOOL 3

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COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

C02-AT3

COURSE OUTCOME COVERED

- Question 1 (5 Marks)

Answer:

Responsive Web Design (RWD) is a web design approach that makes a website automatically adjust its layout, images, text, and other elements according to the screen size and device. It allows a website to work properly on **mobile phones, tablets, laptops, and desktops**.

Importance

- Provides a better user experience on all devices.
- Avoids horizontal scrolling.
- Improves readability and navigation.
- Reduces the need to create separate websites for different devices.
- Improves accessibility and usability.

Key CSS Techniques

1. **Flexible layouts** using %, rem, em, vw, and vh.
2. **Media queries** to apply different styles for different screen sizes.
3. **Responsive images** using max-width: 100%.
4. **Flexbox** for flexible one-dimensional layouts.
5. **CSS Grid** for flexible two-dimensional layouts.

Example using Flexbox

- Question 2 (5 Marks)

Answer:

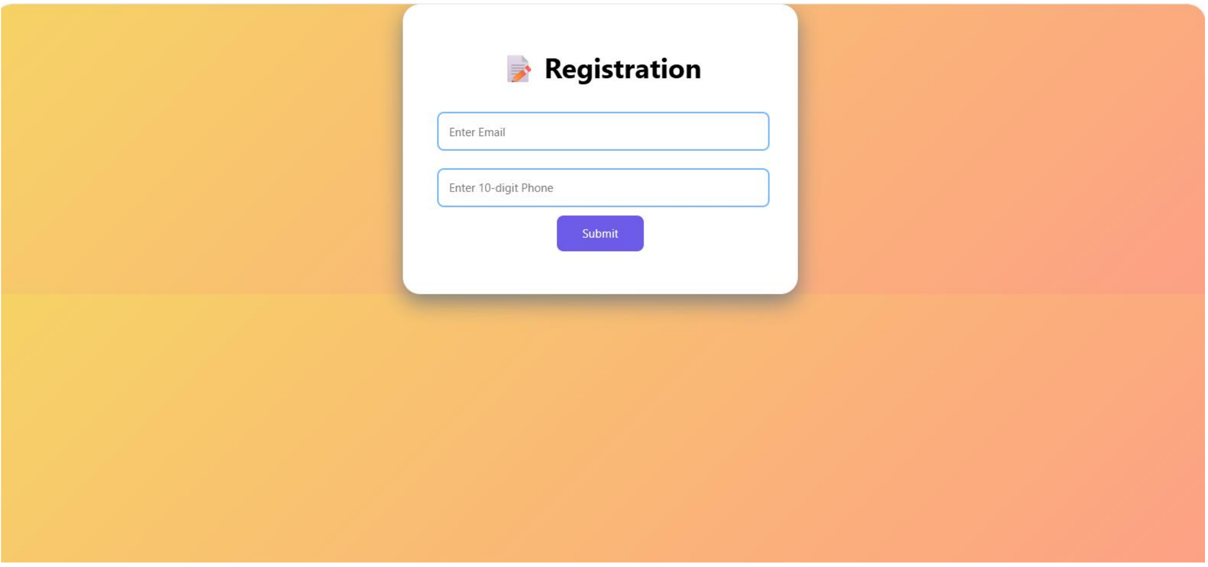
Client-side validation checks user input in the browser before the form is submitted to the server. JavaScript is commonly used for this purpose.

Purpose

- Prevents incorrect data from being submitted.
- Gives immediate feedback to users.
- Reduces unnecessary server requests.
- Improves user experience.

JavaScript Event Handling

JavaScript can use the submit event to validate a form.



Improvements and Security

- Display clear error messages near the input fields.
- Validate required fields.
- Use HTML attributes such as required, type="email", and minlength.
- **Always validate again on the server side**, because client-side validation can be bypassed.
- Sanitize and safely handle submitted data.

➢ Question 3 (5 Marks)

Answer:

The **CSS Box Model** describes how every HTML element is represented as a rectangular box.

The four components are:

1. **Content** – actual text/image.
2. **Padding** – space around the content.
3. **Border** – boundary around padding.
4. **Margin** – space outside the element.

CSS Positioning

Common positioning types are:

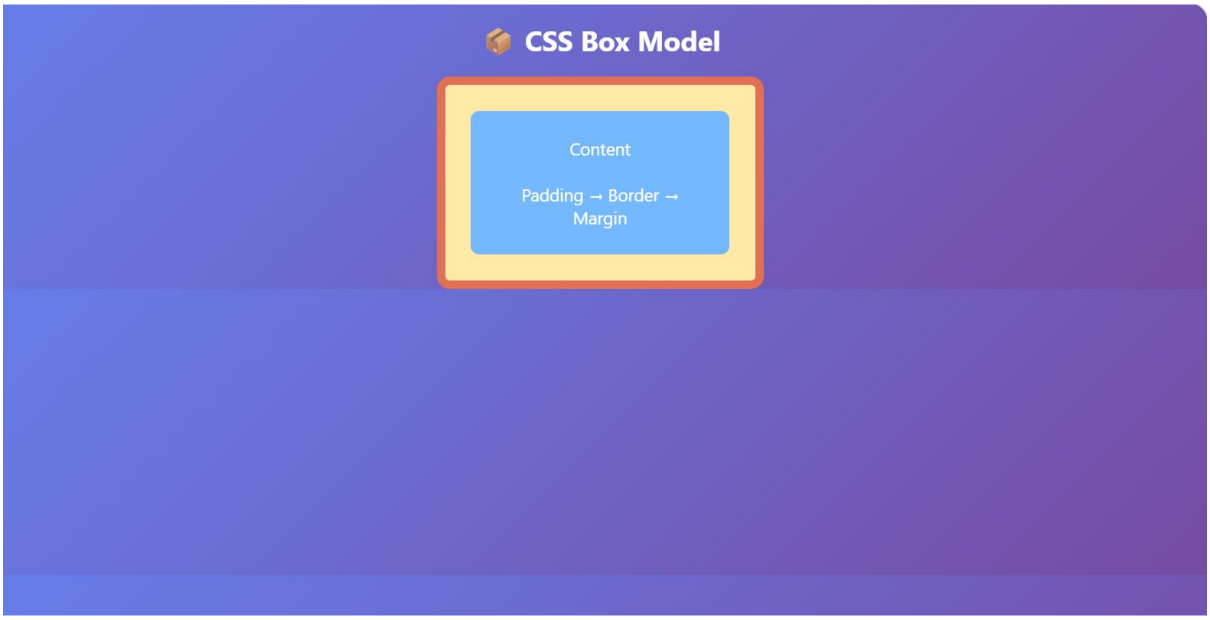
- static – normal document flow.
- relative – moves relative to its normal position.

- absolute – positioned relative to a positioned ancestor.
- fixed – remains fixed relative to the browser window.
- sticky – behaves like relative until a scrolling position is reached.

Causes of Layout Breaking

- Fixed pixel widths.
- Excessive margins and padding.
- Large images without responsive sizing.
- Incorrect use of position: absolute.
- Elements overflowing their containers.
- Lack of media queries.
- Different screen sizes.

Media Query Solution



Best Practices

- Prefer flexible units such as %, rem, and vw.
- Use Flexbox or Grid.
- Use max-width: 100% for images.
- Avoid unnecessary absolute positioning.
- Use media queries for different screen sizes.
- Test the page on mobile, tablet, and desktop.

➢ Question 4 (5 Marks)

Answer:

The **DOM (Document Object Model)** is a programming representation of an HTML document. It represents the webpage as a **tree of objects**.

DOM Manipulation

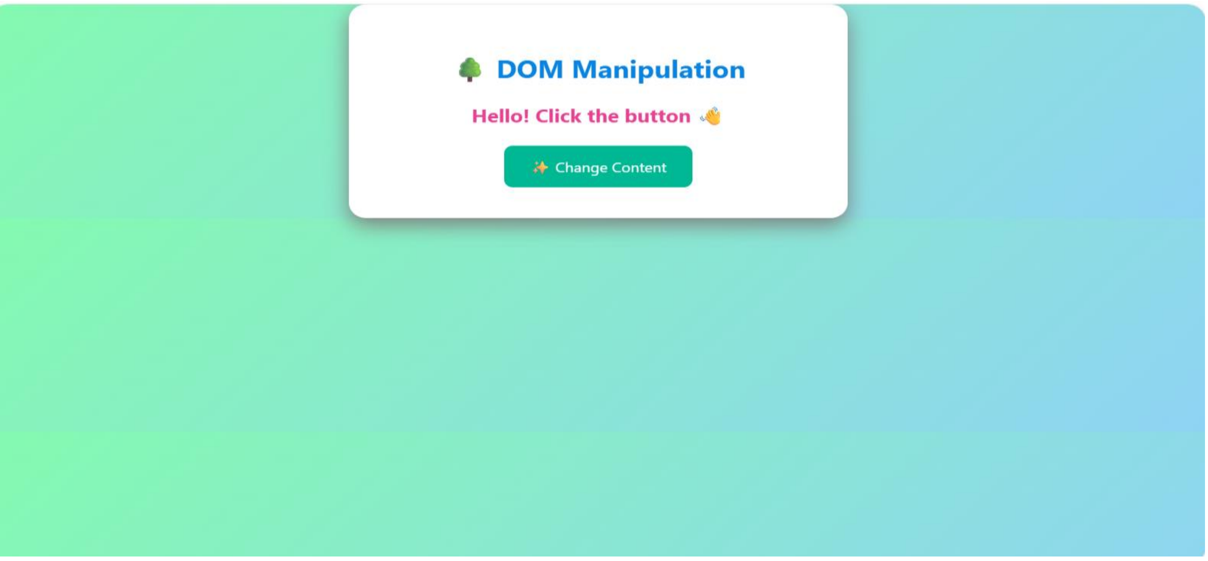
JavaScript can:

- Change text.
- Change styles.
- Add or remove elements.
- Change attributes.
- Respond to user actions.

Improvements

- Use addEventListener() instead of inline event handlers for cleaner code.
- Update only the required DOM elements.
- Avoid unnecessary DOM operations.
- Use efficient event handling.
- Use asynchronous techniques such as fetch() when data must be obtained from a server.
- Provide loading indicators and error messages for better user experience.

Example



Code of Conduct:

I ____________________ (P. Sravanthi / 192311272) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sravanthi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand